

Nyquist

Script: Chapter 5 (Sampling Theorem, Aliasing, Reconstruction)

Learning goals

- Predict alias frequencies with the folding rule and the folding diagram; determine minimum sampling rates, including band-pass (under-)sampling.
- Observe aliasing of the knock resonances on real (simulated) microcontrollers and verify the predictions quantitatively.
- Compare zero-order hold, linear and sinc reconstruction.
- Design an anti-aliasing filter: trade-off between filter order, sampling rate and passband loss.

Background

Sampling with $f_s = 1/T_s$ makes all frequencies $f + kf_s$, $k \in \mathbb{Z}$, indistinguishable. A real tone of frequency f therefore appears at the *alias frequency*

$$f_a = |f - kf_s|, \quad k = \text{round}(f/f_s), \quad 0 \leq f_a \leq f_s/2.$$

A band-limited signal ($X(f) = 0$ for $|f| \geq B$) is uniquely determined by its samples if $f_s > 2B$ and can be reconstructed by sinc interpolation $x(t) = \sum_n x[n] \text{sinc}((t - nT_s)/T_s)$, $\text{sinc}(u) = \sin(\pi u)/(\pi u)$. Everything above $f_s/2$ has to be removed *before* sampling by an analog anti-aliasing (AA) filter.

In our virtual engine the knock sensor has resonances at 6.5 kHz and 10.5 kHz ($\tau = 0.8$ ms) and valve impacts at 8 kHz ($\tau = 0.4$ ms). The **engine-sim** chip updates its analog outputs every 20 μ s (attribute `fs_khz = 50`) and holds the value in between: the chip itself is a sampled system with zero-order hold.

Exercise 5.1 Aliasing by hand

PAPER

- Compute the alias frequencies of the components 6.5 kHz, 8 kHz, 10.5 kHz and of a (hypothetical) third knock mode at 15 kHz for $f_s = 5, 8.33, 16.67, 25$ and 50 kHz. (8.33 kHz is close to the maximum rate of the Uno ADC.)
- Draw the folding diagram (f_a over f , 0 kHz to 16 kHz) for $f_s = 8.33$ kHz and mark the four components. Which two components can no longer be separated after sampling?
- What is the minimum sampling rate for a knock signal with components up to 15 kHz? A 4th-order Butterworth AA filter with $f_c = 15$ kHz ($|H| = 1/\sqrt{1 + (f/f_c)^8}$) is used. Which f_s is needed so that every frequency that folds into 0 kHz to 15 kHz is attenuated by at least 40 dB?
- Band-pass sampling: an analog band-pass passes only the first knock mode, 6 kHz to 7 kHz. Which sampling rates are allowed ($2f_H/k \leq f_s \leq 2f_L/(k-1)$, $k = 1, 2, \dots$)? Show that the Uno at 5 kHz could legally capture this band. Where does the band appear, and is it spectrally inverted?

Exercise 5.2 Aliasing made visible

WOKWI

Two projects record the **KNOCK** output of the engine-sim chip (knock probability 100 %, 3000 rpm) together with the ground-truth outputs **KNK** (HIGH during a knock burst) and **TDC**, and print blocks of CSV lines `n,adc,knk,tdc`: `code/wokwi/sheet05_alias_uno/` (Uno, Timer1 ISR, 512 samples per block) and `code/wokwi/sheet05_alias_esp32/` (ESP32, deadline loop, 2500 samples

per block). The JSLinux program `code/jslinux/sheet05/alias_scan.c` cuts out every knock burst (from the rising KNK edge, 4τ) and every valve burst (TDC $+90^\circ$, 5τ), averages the spectra $|X(f)|^2$ of the segments and lists the peaks.

- Uno: the sampling period is $TS_US = DIV \times 20\mu s$ (default $DIV = 10$, 5 kHz). Compute OCR1A (prescaler 8) and complete the ISR (store KNK in bit 14, TDC in bit 15). Why does the sketch disable the Timer0 interrupt during acquisition?
- Copy at least three blocks from the Serial Monitor into `uno.csv`, implement `alias_freq()` and the start index of the valve burst in `alias_scan.c` and run `./alias_scan uno.csv 5000 10`. Compare the spectral peaks with Exercise 5.1. Look at one knock burst in the CSV: how many oscillation periods do you see, and of which frequency? Repeat with $DIV = 6$ (8.33 kHz). Check with the logic analyzer that the sample marker (D0) is still regular.
- ESP32: complete the deadline loop ($TS_US = 40$, 25 kHz) and verify that no sample is late. Analyse with `./alias_scan esp.csv 25000 12`. Then set $TS_US = 60$ (16.67 kHz). Why is this particularly dangerous for a knock detector that integrates the energy in a band around 6.5 kHz?
- Why must the sampling period be a multiple of $20\mu s$? Consider the Uno at exactly 8 kHz ($125\mu s$): determine the sequence of time offsets between the ADC instants and the last chip update, its effect on the SNR (Exercise 4.1 e) and a value of the chip attribute `fs_khz` in `diagram.json` that fixes the problem.
- `./alias_scan sim 5000 10` simulates the chip and the ADC (“virtual Wokwi”, also writes `alias_sim.csv`). Compare its peaks with your Wokwi measurement.

The UART transmits about 11.5 characters per ms: printing a CSV line per sample is impossible at kHz rates, therefore samples are buffered in RAM (Uno: 512 samples = 1 KB of its 2 KB). Wokwi is not cycle-accurate for the ESP32 – if `late` is not 0, increase TS_US to the next multiple of 20.

Exercise 5.3 Reconstruction and anti-aliasing

JSLINUX

Reconstruction. `code/jslinux/sheet05/recon.c` represents a “continuous” sine on a grid of $M = 40$ points per sampling interval ($f_s = 25$ kHz, 600 samples) and computes the normalised error $NMSE = \sum e^2 / \sum x^2$ of a reconstruction on this grid.

- Implement zero-order hold, linear interpolation and sinc interpolation with $\pm K = 16$ terms, once plain (truncated) and once with a Hann taper $w(u) = \frac{1}{2} + \frac{1}{2} \cos(\pi u / (K + 1))$. Show analytically that the ZOH of a sine has $NMSE = 2(1 - \sin(\omega T_s) / (\omega T_s))$ and compare. Discuss the table (method vs. f/f_s).
- A 15 kHz tone is sampled at 25 kHz and sinc-reconstructed. What comes out? Predict frequency and phase.

Anti-aliasing. `code/jslinux/sheet05/antialias.c` simulates the analog world on a 1 MHz grid; the AA filter (cut-off $f_c = 12$ kHz) is simulated on this grid, the ADC takes every $(1 \text{ MHz} / f_s)$ -th value. Test signals: knock tones 6.5 kHz/10.5 kHz; a disturbance at $f_d = f_s - 6.5$ kHz that aliases exactly onto the knock frequency; white noise (bandwidth 500 kHz).

- Implement the RC low-pass (exact update for an input held constant over Δt), the 4th-order Butterworth as a cascade of two 2nd-order sections ($Q_k = 1 / (2 \sin(\frac{(2k-1)\pi}{2n}))$, given biquad simulator) and the amplitude measurement at a given frequency in the sampled signal. Run the program for $f_s = 25$ and 50 kHz and filter orders 0, 1, 2, 4. Compare the alias amplitudes with $|H(f_d)|$.
- Which Butterworth order would be required for 40 dB alias suppression in the band 0 kHz to 10.5 kHz with $f_c = 12$ kHz at $f_s = 25$ kHz and at 50 kHz? Formulate a design recommendation

for the knock ADC.

Deliverables

5.1: alias table, folding diagram, AA and band-pass sampling calculation. 5.2: completed sketches, CSV files of the Uno (two rates) and the ESP32 (two rates), output of `alias_scan` with comparison to 5.1, answers to c) and d). 5.3: completed programs, both tables with discussion, filter-order recommendation.